SHYAMGANESH S

AI Engineer | Computer Vision & Generative AI | MSc AI & ML – University of Birmingham Dubai (Sept 2026)
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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience across the full computer vision and deep learning pipeline — data preprocessing, model training, hyperparameter tuning, fine-tuning, and deployment. Built real-time YOLO-based object detection systems for industrial inspection and a multi-class CNN-based medical imaging platform with LLM-powered report generation. Published IEEE researcher in deep learning for medical diagnostics. Incoming MSc Artificial Intelligence & Machine Learning student at the University of Birmingham Dubai (Sept 2026), targeting AI Engineer, Machine Learning Engineer, and Computer Vision Engineer roles in Dubai's technology sector.

TECHNICAL SKILLS

Programming: Python, SQL, Object-Oriented Programming (OOP), Data Structures & Algorithms

Machine Learning & Deep Learning: TensorFlow, Scikit-learn, Model Training & Evaluation, Hyperparameter Tuning, Feature Engineering, Transfer Learning, Data Preprocessing

Computer Vision: YOLO (Object Detection & Tracking), OpenCV, CNN Architectures, Image Preprocessing (CLAHE)

Generative AI & NLP: Large Language Models (LLMs), LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Prompt Engineering, Natural Language Processing (NLP)

MLOps & Deployment: Docker, Kubernetes (Basic), CI/CD, Model Deployment & Monitoring, FastAPI, Flask, REST APIs

Cloud & Tools: AWS Cloud, Git & GitHub

PROFESSIONAL EXPERIENCE

ML / Data Science Intern — Seewise.ai *Mar 2026 – Jul 2026*

- Engineered real-time computer vision pipelines for industrial inspection using YOLO-based object detection and tracking.
- Designed and executed dataset preparation and preprocessing workflows; improved model accuracy through systematic hyperparameter tuning.
- Fine-tuned detection models and supported end-to-end deployment and client integration workflows.

Python Development Intern — HCL Technologies Ltd. *Oct 2024 – Apr 2025*

- Built Python automation scripts using OOP, file handling, and scripting for internal training modules.
- Applied Git version control and industry coding standards in a collaborative software development environment.

KEY PROJECTS

MedLens AI — Chest X-Ray Analysis Platform

- Built a chest X-ray analysis platform, expanding a single-class YOLO detector into 14-class multi-disease detection.
- Designed OpenCV (CLAHE)-based image preprocessing and a structured diagnostic output pipeline for clinical interpretability.
- Architected a cost-optimized, multi-provider LLM failover system for automated report generation on CPU-only deployment.

AI-Based Water Bottle Inspection System

- Built a real-time defect detection system using YOLO with object tracking and unique ID assignment.
- Optimized the inference pipeline to achieve production-grade real-time throughput.

Disease Prediction Systems (Potato & Parkinson's)

- Trained CNN classifiers with transfer learning and data augmentation for plant disease detection.
- Built ML models on structured medical data with feature engineering; co-authored IEEE IATMSI 2025 paper, "Enhancing Parkinson's Disease Diagnosis: A Deep Learning Framework."

EDUCATION

MSc Artificial Intelligence & Machine Learning — University of Birmingham Dubai *Sept 2026*

B.E. Computer Science & Engineering — Kongu Engineering College *2025*

CERTIFICATIONS & ACHIEVEMENTS

- IBM Data Science Certification — Coursera
- IBM Data Science with AI Certification — SLA Institute
- Machine Learning & AI Foundations — Self-Paced Training
- National Gold Medalist, RSFI Roller Skating (Vizag); National Silver (SGFI Delhi) & Bronze (SGFI Mumbai)
- Treasurer, CSI | Executive Member, App Dev Club | House Captain